

SAT UZBEKISTAN

Equations

Test creator Xodiyev Lutfulloxon

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➤ Easy

1. If $x = y$ and $x + y = 10$, then $2x + y =$
(A) 3
(B) 15
(C) 18
(D) 24
(E) 30
2. If $2x + 1 = 3x + 2$, then $5x + 2 =$
(A) -5
(B) -3
(C) -1
(D) 0
(E) 3
3. If $7x + 3y = 12$ and $3x + 7y = 8$, then $x - y =$
(A) 1
(B) 3
(C) 7
(D) 8
(E) 12
4. If $4p$ is equal to $6q$, then $2p - 3q$ equals which one of the following?
(A) 0
(B) 2
(C) 3
(D) 4
(E) 6
5. Which one of the following must equal $p + q$, if $x - y = p$ and $2x + 3y = q$?
(A) $x + y$
(B) $3x - 2y$
(C) $2x - 3y$
(D) $2x + 3y$
(E) $3x + 2y$
6. If $l + t = 4$ and $l + 3t = 9$, then which one of the following equals $l + 2t$?
(A) $13/2$
(B) $19/2$
(C) $15/2$
(D) $17/3$
(E) $21/4$
7. If $(x - y)(x + y) = 15$ and $x + y = 5$, then what is the value of x/y ?
(A) 3
(B) 4
(C) 5
(D) 10
(E) 15

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8. If $x - 4y = 1$ and $y = x/2 + 1$, then what is the value of x ?

- (A) -5
- (B) -2
- (C) 2
- (D) 5
- (E) 8

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➤ **Medium**

9. If $x^2 - 4x + 3$ equals 0, then what is the value of $(x - 2)^2$?

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) 4

10. If $x = a + 2$ and $b = x + 1$, then which one of the following must be true?

- (A) $a > b$
- (B) $a < b$
- (C) $a = b$
- (D) $a = b^2$
- (E) $a = b^3$

11. If p is the sum of q and r , then which one of the following must equal $q - r$?

- (A) $p - r$
- (B) $p + r$
- (C) $p - 2r$
- (D) $p + 2r$
- (E) $2q - p$

12. The sum of two numbers is 13, and their product is 30. What is the sum of the squares of the two numbers?

- (A) -229
- (B) -109
- (C) 139
- (D) 109
- (E) 229

13. If $p + q = 7$ and $pq = 12$, then what is the value of $\frac{1}{p^2} + \frac{1}{q^2}$?

- (A) $1/6$
- (B) $25/144$
- (C) $49/144$
- (D) $7/12$
- (E) $73/144$

14. If $2x + 3y = 11$ and $3x + 2y = 9$, then $x + y =$

- (A) 4
- (B) 7
- (C) 8
- (D) 9
- (E) 11

15. If $2x = 2y + 1$, then which one of the following is true?

- (A) $x > y$
- (B) $x < y$
- (C) $x = y$
- (D) $x = y + 1$
- (E) $y = x + 1/2$

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16. If $3x + y = x + 2y$, then $2x - y =$

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) 4

17. If $yz - zx = 3$ and $zx - xy = 4$, then $xy - yz =$

- (A) -7
- (B) 1
- (C) 3
- (D) 4
- (E) 7

18. If $(x + 5) \div \left(\frac{1}{x} + \frac{1}{5}\right) = 5$, then $x =$

- (A) -5
- (B) $1/2$
- (C) 1
- (D) 5
- (E) 10

19. If $(x + 5)\left(\frac{1}{x} + \frac{1}{5}\right) = 4$, then $x =$

- (A) $1/5$
- (B) $1/2$
- (C) 1
- (D) 5
- (E) 10

20. If $\frac{a^2 - 9}{12a} = \frac{a - 3}{a + 3}$, $a + 3 \neq 0$, and $a \neq 0$, then $a =$

- (A) 1
- (B) 2
- (C) 3
- (D) 6
- (E) 9

21. If both expressions $x^2 - 3x + 2$ and $x^2 - 4x + 3$ equal 0, then what is the value of $(x - 3)^2$?

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) 4

22. If $42.42 = k(14 + 7/50)$, then what is the value of k ?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

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23. If $|2x - 4|$ is equal to 2 and $(x - 3)^2$ is equal to 4, then what is the value of x ?

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

24. If $(a + 2)(a - 3)(a + 4) = 0$ and $a > 0$, then $a =$

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

► **Hard**

25. If $x + y = 7$ and $x^2 + y^2 = 25$, then which one of the following equals the value of $x^3 + y^3$?

- (A) 7
- (B) 25
- (C) 35
- (D) 65
- (E) 91

26. A system of equations is as shown below

$$\begin{aligned}x + l &= 6 \\x - m &= 5 \\x + p &= 4 \\x - q &= 3\end{aligned}$$

What is the value of $l + m + p + q$?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

27. If $mn = 3$ and $1/m + 1/n = 4/3$, then what is the value of $0.1 + 0.1^{1/m} + 0.1^{1/n}$?

- (A) $0.2 + 0.1^{1/3}$
- (B) $0.1 + 0.1^{1/3} + 0.1^{1/2}$
- (C) $0.1 + 0.1^{4/3} + 0.1^{1/2}$
- (D) $0.1 + 0.1^{1/3} + 0.1^{3/2}$
- (E) $0.1 + 0.1^{1/4} + 0.1^{1/2}$

28. If $(x-2y)(x+2y) = 5$ and $(2x-y)(2x+y) = 35$, then $\frac{x^2-y^2}{x^2+y^2} =$

- (A) $-8/5$
- (B) $-4/5$
- (C) 0
- (D) $4/5$
- (E) $7/5$

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29. $a, b,$ and c are three different numbers. None of the numbers equals the average of the other two. If $\frac{x}{a+b-2c} = \frac{y}{b+c-2a} = \frac{z}{c+a-2b}$, then $x + y + z =$

- (A) 0
- (B) 3
- (C) 4
- (D) 5
- (E) 6

30. If $\frac{l}{m+n} = \frac{m}{n+l} = \frac{n}{l+m} = k$, where k is a real number, then which one of the following does k equal?

- (A) $1/3$
- (B) $1/2$
- (C) 1
- (D) 2
- (E) 3

31. If $a, b,$ and c are three different numbers and $\frac{x}{b-c} = \frac{y}{c-a} = \frac{z}{a-b}$, then what is the value of $ax + by + cz$?

- (A) 0
- (B) 1
- (C) 2
- (D) 3
- (E) 4

32. $a, b,$ and c are three different numbers, none of which equals the average of the other two. If $\frac{x}{b-c} = \frac{y}{c-a} = \frac{z}{a-b}$, then $x + y + z =$

- (A) 0
- (B) $1/2$
- (C) $1/3$
- (D) $2/3$
- (E) $3/4$

33. If $x + y = 750$, then which one of the following additional details will determine the value of x ?

- (A) $x + 2y = d$
- (B) $2x + 4y = 2d$
- (C) $2x + 2y = 1500$
- (D) $3x = 2250 - 3y$
- (E) $2x + y = 15$

➤ **Very Hard**

34. If a , b , and c are not equal to 0 or 1 and if $a^x = b$, $b^y = c$, and $c^z = a$, then $xyz =$

- (A) 0
- (B) 1
- (C) 2
- (D) a
- (E) abc

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